How Do You Discover a New Number?

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1 Introduction

Most of us are somewhat familiar with number systems differing from our default. Whether that is the prototypical complex number system, or something slightly more advanced, like the quaternions that I have previously discussed, an example of a hypercomplex number system.

I want to do something a little bit different in this document, I would like to discuss how I came across a number system that was previously unknown to me, and how I would encourage you, the reader, to try the same thing.

2 Setting the Stage

As you may have noticed from my recent documents, I have been enjoying group theory. It provides a nice systematic way to find matrix representations of objects that we may not usually think of in terms of matrices, like imaginary numbers. For example, we consider the isomorphism $\mathrm{U}(1) \cong \mathrm{SO}(2)$. We recall that the elements of the former group are unitary complex numbers, and the latter are rotation matrices. Following from the definition of a Lie algebra, the basis vector for $\mathfrak{u}(1)$ should be i. Doing the same for $\mathfrak{so}(2)$ will yield a matrix representation of i. We can find this to be:

$$i \Longleftrightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{1}$$

We can find that this matrix behaves just as we would expect i to behave. Try it yourself! It's very entertaining.

3 Next Steps

Following from this, I wanted to repeat a similar process for the group $\mathrm{SO}(1,1)$. At the time, I considered the hyperbolic analogue of Euler's identity:

$$e^{x} = \cosh(x) + \sinh(x) \tag{2}$$

I assumed that this statement described a hyperbolic rotation, just as its complex sibling describes circular rotations. We can see that the number 1 takes the place of i. So, before seeking the basis vector for $\mathfrak{so}(1,1)$, I assumed it would just be the 2×2 identity matrix, $\mathbb{I}_2$.

However, when I worked out this basis vector J, I obtained:

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{3}$$

crowd gasping It's backwards! Naturally, I assumed I had made a mistake, but upon realising I hadn't, I just rolled with it, treating it as if it was just the number 1. However, when I squared it, something unusual happened:

$$J^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{4}$$

It squared to the identity, to 1! So it could not itself be 1. This was very unusual, to me, no such number existed. To me, there did not exist an x such that:

$$\{x : x \neq 1,\, x^2 = 1\} \tag{5}$$

I spoke to a friend, and we found this to be a matrix representation of what was called the "split complex numbers" or "dual complex numbers". I had never heard of it before. But these numbers, J, form a true hyperbolic Euler identity as:

$$e^{Jx} = \cosh(x) + J \sinh(x) \tag{6}$$

You can in fact verify this yourself, by exponentiating J in matrix form, one obtains a hyperbolic rotation matrix (as one would expect, given that it is an element of $\mathfrak{so}(1,1)$.

4 What Next?

Now, admittedly, this isn't very useful. In general, these wacky number systems aren't that useful just because it is often much easier to just work with the matrix itself. But the process of trivial experimentation that led to this was very eye opening, I must say, and I'd certainly encourage you to test these more simple thing to see if you too can come across something you may not expect. The beauty of mathematics is that it is chock-full of these glistening specks of beauty, you just have to find them!